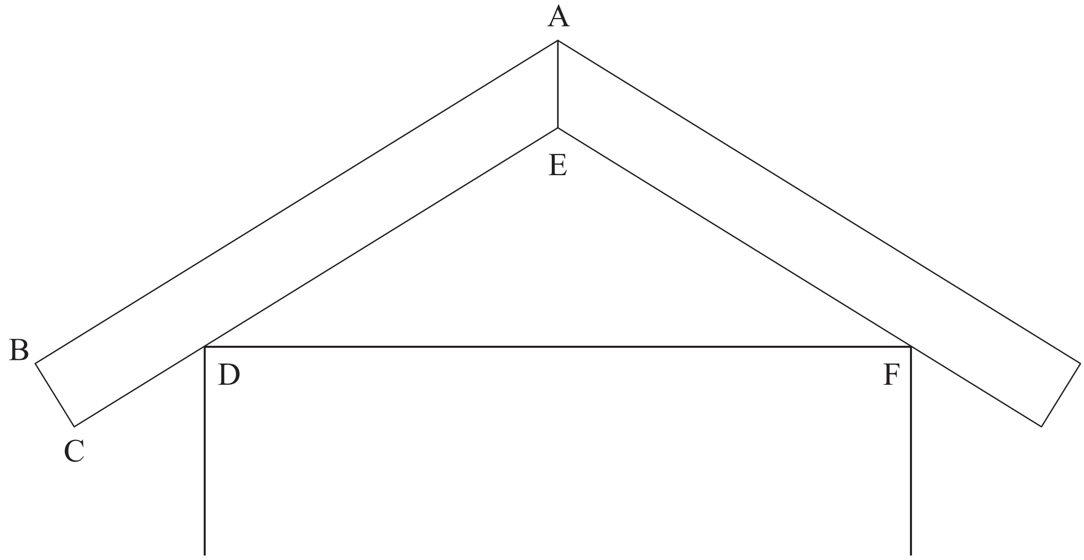
4. [Maximum mark: 8]

The following diagram shows the cross section of the roof of a house. The cross section is symmetrical about the vertical line through points A and E.

diagram not to scale



The gradient of $[\mathrm{BA}]$ is $\dfrac{7}{12}$.

(a) Find the size of $\mathrm{B\hat{A}E}$, expressing your answer in degrees. [3]

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(This question continues on the following page)

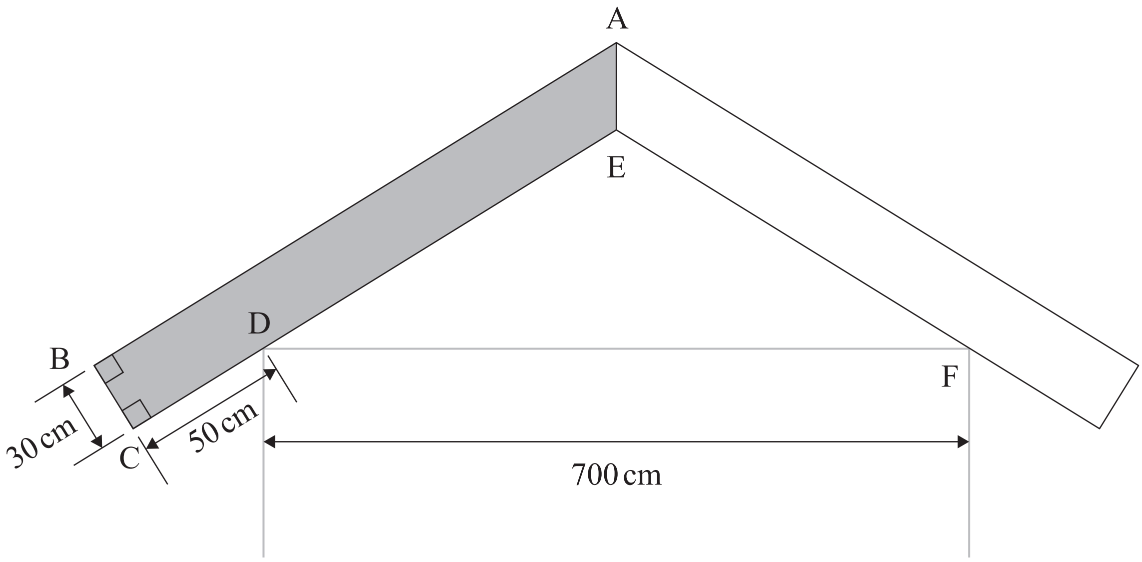
(Question 4 continued)

A builder requires the lengths of the sides $[BA]$ and $[CE]$.

The builder has the following measurements:

$A\hat{B}C = B\hat{C}E = 90°$, $DC = 50\,cm$, $BC = 30\,cm$, and $DF = 700\,cm$.

diagram not to scale



(b) Find

(i) CE;

(ii) BA. [5]

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